

Simpler Model Link Seed and Growth Directly

Data

We now have a time-series across individuals $i = 1 \dots, I$ in year $t = 1, \dots, T$:

- $sc_{i,t} \in \mathbb{N}$: seed count of tree i in year t
- $BAI_{i,t} \geq 0$: basal area increment of tree i in year t
- $DBH_{i,t} \geq 0$: DBH of tree i in year t
- $GST_t \in \mathbb{R}$: growing season temperature in year t

Parameters

Growth parameters

- $\alpha_{BAI} \in \mathbb{R}$: intercept for growth
- $\beta_{GST1} \in \mathbb{R}$: effect of climate on growth
- $\sigma_{BAI} > 0$: standard deviation for growth

Reproduction parameters

- $\alpha_{sc} \in \mathbb{R}$: intercept for seeds
- $\beta_{dbh} \in \mathbb{R}$: effect of tree size on seeds
- $\beta_{GST2} \in \mathbb{R}$: effect of climate on seeds
- $\gamma_{current} \in \mathbb{R}$: current year's growth effect on seeds
- $\gamma_{lag} \in \mathbb{R}$: lag effect, last year's growth effect on seeds
- $\phi_{sc} > 0$: dispersion parameter

Prior Distributions

$$\alpha_{\text{BAI}} \sim \mathcal{N}(0, 5)$$

$$\beta_{\text{GST1}} \sim \mathcal{N}(0, 5)$$

$$\sigma_{\text{BAI}} \sim \mathcal{N}(0, 1)$$

$$\alpha_{\text{sc}} \sim \mathcal{N}(0, 5)$$

$$\beta_{\text{dbh}} \sim \mathcal{N}(0, 5)$$

$$\beta_{\text{GST2}} \sim \mathcal{N}(0, 5)$$

$$\phi_{\text{sc}} \sim \text{Gamma}(2, 0.1)$$

$$\gamma_{\text{current}} \sim \mathcal{N}(0, 5)$$

$$\gamma_{\text{lag}} \sim \mathcal{N}(0, 5)$$

Model

For each individual tree i and year t :

Expected log-growth for all $t = 1, \dots, T$:

$$G_{i,t} = \alpha_{\text{BAI}} + \beta_{\text{GST1}} \cdot \text{GST}_t$$

Observed BAI:

$$\text{BAI}_{i,t} \sim \text{LogNormal}(G_{i,t}, \sigma_{\text{BAI}})$$

Expected seed count for all $t = 1, \dots, T$:

$$\log(\mu_{\text{sc},i,t}) = \alpha_{\text{sc}} + \beta_{\text{dbh2}} \cdot \text{DBH}_{i,t} + \beta_{\text{GST2}} \cdot \text{GST}_t + \gamma_{\text{current}} \cdot G_{i,t} + \gamma_{\text{lag}} \cdot G_{i,t-1}$$

Observed seed counts:

$$sc_{i,t} \sim \text{NegBinomial2Log}(\log(\mu_{\text{sc},i,t}), \phi_{\text{sc}})$$